POLYMERS

Introduction

A polymer is a large molecule made by linking together repeating units of small molecules (called monomers)

A polymer always consists of hundreds to thousands of monomer units. So they have very high molecular masses.

All polymers are macromolecules but all macromolecules are not polymers.

⇒ **Homopolymer:-** Polymers which are formed by only one type of monomer units, called homopolymers.

⇒ **Copolymers:-** Polymers which are formed by more than one type of monomers, known copolymers.

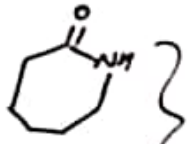
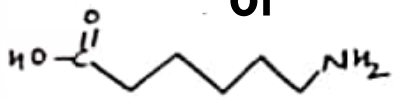
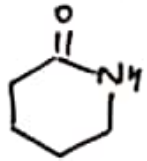
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Examples of Homomers:-

Monomer		Polymer
① α-D-Glucose	⇒	Starch
② β-D-Glucose	⇒	Cellulose
③ α-D-Glucose	⇒	Glycogene
④ Fructose	⇒	Inuline
⑤ $CH_2=CH_2$	⇒	Polyether (Polythene)
⑥ $CH_2=CH-Cl$ (vinyl chloride)	⇒	PVC / Poly vinyl chloride
⑦ $CH_3-\overset{O}{\overset{\|}{C}}-O-CH=CH_2$ (vinyl acetate)	⇒	PVA (Poly vinyl acetate)

Monomer		Polymer
⑧ ε caprolactam  or  6-aminohexanoic acid	⇒	Nylon-6 / Perlon-L
⑨ δ-Valarolactam 	⇒	Nylon-5
⑩ $CH_2=CH-CN$ (acrylonitrile)	⟹	{ PAN / Polyacrylonitrile / Orlon / acrilane }
⑪ $CH_2=CH$ | Ph Styrene	⟹	(Polystyrene / Styron

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Examples of Homomers:-

Monomer	Polymer
(12) $F_2C=CF_2$ Tetra fluoro ethelene	Teflon/PTFE
(13) $CH_2=\overset{Me}{\overset{|}{C}}-\overset{O}{\overset{\|\|}{C}}-O-Me$ Methyl methacrylate	PMMA

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Examples of copolymer

monomers

Polymer

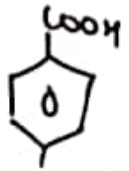
① $H_2N(CH_2)_6NH_2$ + $HO-\overset{O}{\overset{\|}{C}}-(CH_2)_4-\overset{O}{\overset{\|}{C}}-OH$ ⟹ Nylon-6,6

Hexamethylene diamine

Adipic acid

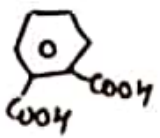
② " + $HO-\overset{O}{\overset{\|}{C}}-(CH_2)_8-\overset{O}{\overset{\|}{C}}-OH$ → Nylon. 6,10

Sabacic acid

③  + $HO-CH_2-CH_2-OH$ → Terelene (Dacron)

Terephalic acid

Glycol

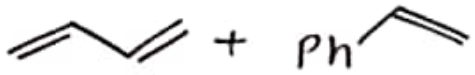
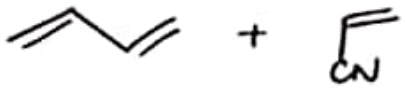
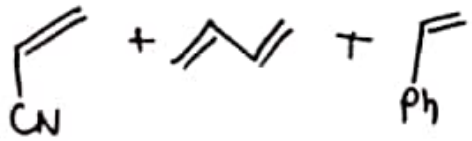
④  + " " ⟹ Glyptal

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Examples of copolymer:

monomers		Polymer
(5)		⇒ Buna-S
(6)		→ Buna-N
		⇒ ABS
(7)		

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POLYMERS

Classification of Polymers:-

[A] On the basis of source:-

(i) Natural polymer

- Which are found in nature (in animal and plants)

Examples

(i) Starch
(ii) Cellulose
(iii) Proteine
(iv) Nucleic acid
(v) Silk
(vi) natural rubber etc
(VII) glycogen

(ii) Synthetic polymer (man made polymer)

Which are prepared in laboratory

Examples

① Polythene ⑥ PAN
② PVC ⑦ PVA
③ Nylons ⑧ Teflon etc.
④ Neoprene
⑤ Terelene

(iii) Semi Synthetic Polymer

modified derivatives of natural polymer

Examples

① Cellulose acetate (Rayon)
② Cellulose nitrate etc.

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POLYMERS

CLASSFICATION

[B] On the basis of Structure

(i) Linear polymer

The monomers are joind together to form long straight chains of polymer molecules. These chains are stacked over one another to give a well-packed structure.

⇒ High M.P,
⇒ High density
⇒ High tensile / Pulling strength
⇒ Examples = HDPE, Nylon, Polyesters etc

generally fibres



close packing of chains due to strong intermolecular forces like intramolecular H-bonding, ionic attractions etc.

(ii) Branched Chain Polymer

The monomer units not only combine to produce the linear chain (called main chain) but also form branches along the main chain.

⇒ have lower M.P.
⇒ lower density
⇒ low tensile strength.

Examples = LDPE, Amylopectic, Glycosen Valacanised rubber PVC etc



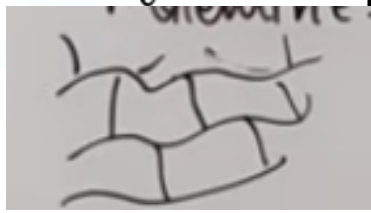
thus, a well-packed structure is difficult

(iii) 3D Polymer or Cross-linked polymer

In these polymers linear polymeric chains are jointed together to form a 3D network structure.

⇒ Hard, rigid and brittle
⇒ These are condensation Polymer. generally

Examples Bakelite, Urea-formaldehyde Polymer, Melamine-formaldehyde Polymer melmac



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POLYMERS

CLASSFICATION

[C] On the basis of molecular forces
- ① Elastomer
- ② Fibers
- ③ Thermoplastic
- ④ Thermosetting polymer

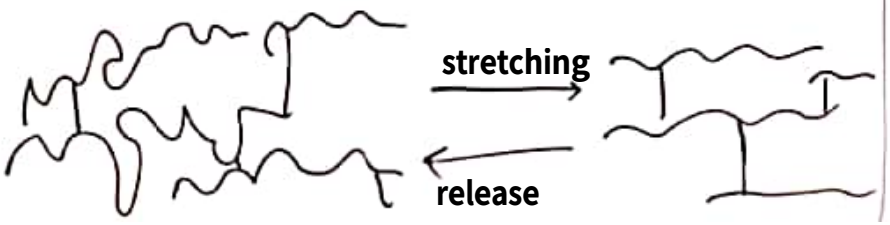
① **Elastomer** :- ⇒ These have weakest force of attraction among polymer chains.

⇒ have high degree of elasticity.

⇒ These polymer consist of randomly coiled molecules

Chains of irregular shapes having a few cross links.

Examples : All Polymer of Butadiene & derivatives of Butadiene (Polybutadiene, Buna-S, Buna-N etc)



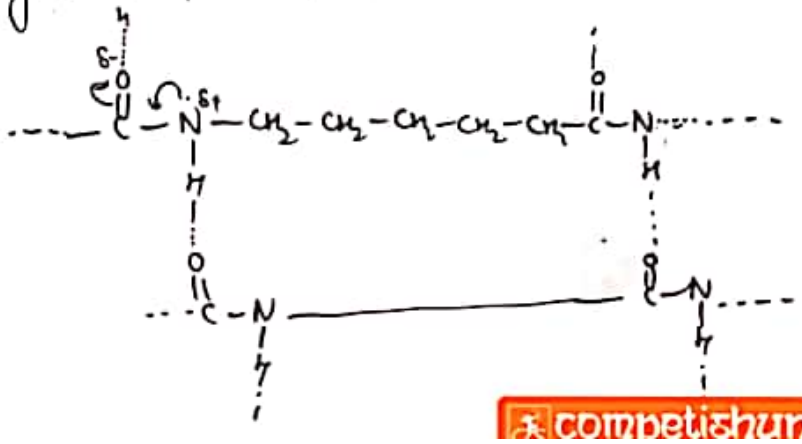
② **Fibers** :-

⇒ Strongest attraction of force between polymer molecules

⇒ These forces are due to either -H-bonding (in Nylons) or dipole-dipole interactions (in polyesters, orlon)

Examples :- Nylon-6 | Nylon-6,6 | Terelene | PAN etc.

⇒ Nylon-6



intermolecular H bonding

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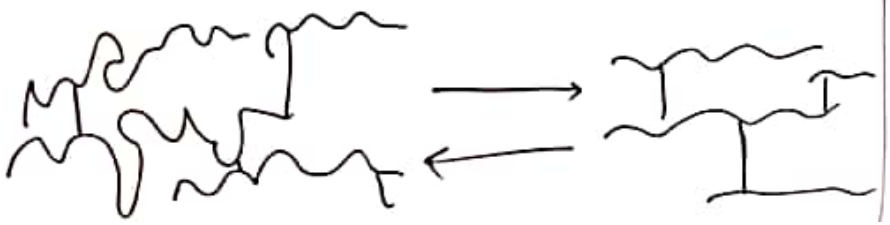
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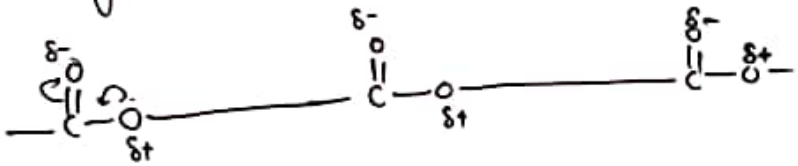
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dipole interactions in polyesters

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③ Thermoplastic

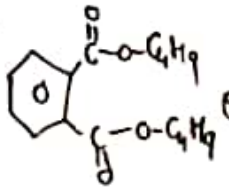
⇒ Attraction forces between elastomer and fibers.

⇒ These are linear polymer which are hard at room temp. but they become soft or viscous on heating. (without change in chemical composition).

→ These plastics can be moulded in different shapes by heating

→ Examples:- Polyethene, Polypropylene, PVC, PVA, PMMA, Buna-S etc.

Plasticisers: Softning agent,

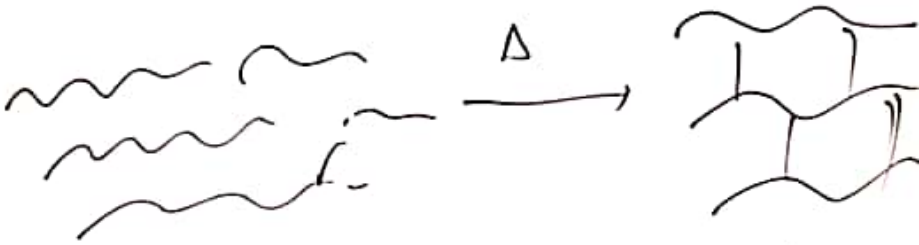


DBP etc. DOP

④ Thermosetting polymer.

- Such types of polymers which can not be moulded are called thermosetting polymer
- In these polymers number of cross links are increased on heating

Examples = Bakelite, Urea formaldehyde resin melmac etc.



competition

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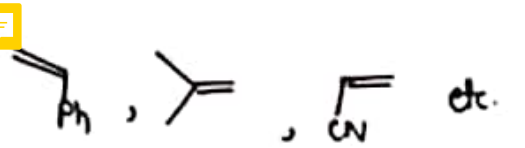
POLYMERS

Classification of Polymers:-

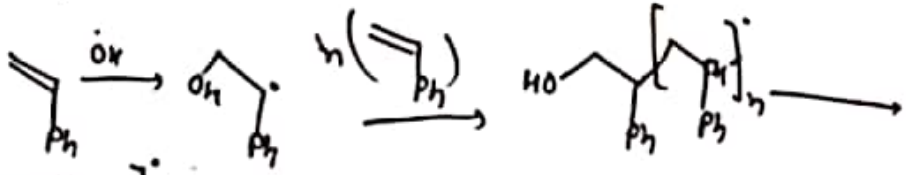
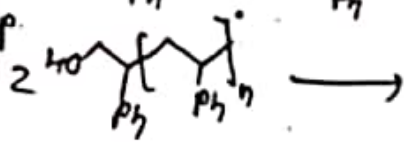
(IV) on the basis of Synthesis :-
- IV.A Addition Polymer (Chain Growth Polymers)
- IV.B condensation polymer (Step Growth polymers)

(IV.A) Addition Polymers: which are formed by addition reactions — usually given by alkenes, alkynes, alkadienes

methods of addition polymerisation:-
- (i) Free radical polymerisation
- (ii) Cationic polymerisation
- (iii) Anionic polymerisation
- (iv) Zeigler Natta catalyst — catalytic polymerisation

(i) Free radical polymerisation:- eg  etc.

monomers whose free radicals are stable use this method of polymerisation

mechanism
- Chain initiation step. $H_2O_2 \longrightarrow 2HO^{\cdot}$
- Chain propagation Step 
- Chain termination step. 

carried out in presence of peroxide/sunlight etc. (free radical initiators)

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POLYMERS

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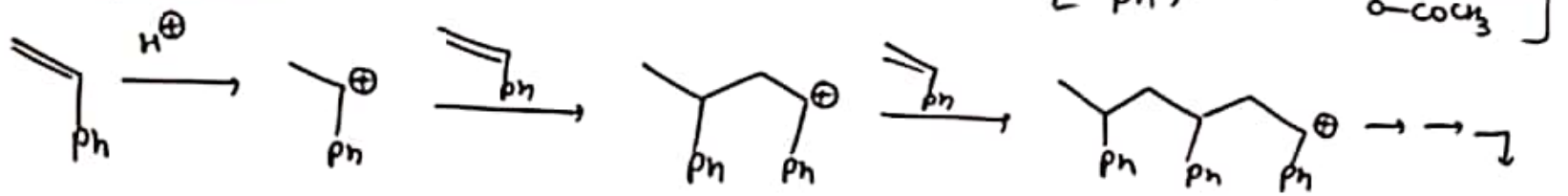
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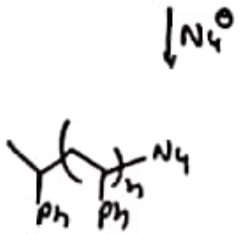
(IV.A) Addition Polymers: which are formed by addition reactions

② Cationic polymerisation:- initiator = $H^{\oplus}$, BF_3 (electrophiles)

ex. [Ph, , $COCH_3$ etc.]



takes place in appropriate temperature, pressure, catalyst conditions



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Classification of Polymers:-

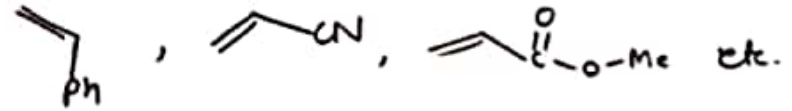
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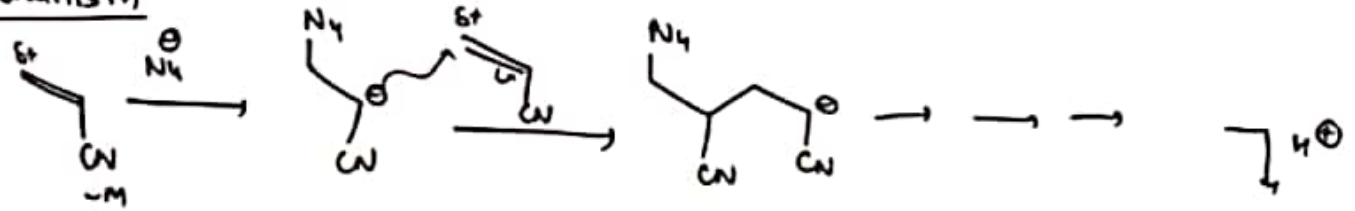
(IV.A) Addition Polymers: which are formed by addition reactions

③ anionic polymerisation:- • initiator $(NH_2^{\ominus}) = H^{\ominus}, \overset{\ominus}{N}H_2$, RLi etc..

given by monomers forming strong carbanion

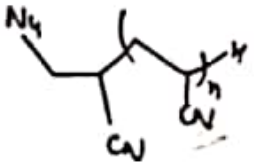
• 

mechanism



formation of PAN

④ Zieler Natta Catalyst = $(TiCl_4 + Et_3Al)$



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POLYMERS

Classification of Polymers:-

(IV) On the basis of Synthesis :-

- IV.A Addition Polymer (Chain Growth Polymers)
- IV.B condensation polymer (Step Growth polymers)

(IV.B) Condensation polymers:- which are form by condensation reaction along with elimination of small groups like water, ammonia, ethyl alcohol etc.

when only two functional grps. involved, formation of straight chain takes place
when more than two functional grps. involved, branching takes place



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Examples of Polymer:-

Addition

A.] Addition Homopolymers

a.) 1-2 addition

① $n CH_2=CH_2 \longrightarrow \{CH_2-CH_2\}_n$ Poly ethylene (Polythene)

- Low Density Poly Ethylene (LDPE)
- High Density Poly Ethylene (HDPE)

$n CH_2=CH_2 \xrightarrow[1000-2000\ atm,\ high\ temp.]{O_2 + Peroxides} LDPE$

trick to learn :
itni mehnat (drastic conditions) lagake bhi LDPE

$n CH_2=CH_2 \xrightarrow[\Delta,\ Pressure\ 4\text{-}5\ atm]{TiCl_4 + AlEt_3} HDPE$

② $CH=CH_2$ (CH3 Propylene) $\xrightarrow{Polymerisation} \{CH(CH_3)-CH_2\}_n$ Polypropylene

higher grade plastic
used in feeding bottles, ropes etc.

③ $CH=CH_2$ (Ph Styrene) $\xrightarrow{\Delta} \{CH(Ph)-CH_2\}_n$ Polystyrene / Styron

used in making of thermocols, hence, polystyrene is used in packing

was used earlier in formation of radio/TV cabinets; now replaced by other polymers

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Examples of Polymer:-

Addition

④ $n\ CH_2=\underset{Cl}{CH}$ — vinyl chloride —Polymerisation→ $\left[-\underset{Cl}{CH}-CH_2-\right]_n$ PVC / Poly vinyl chloride

⑤ $n\ CF_2=CF_2$ — Tetra fluoro ethylene —:→ $\{CF_2-CF_2\}_n$ Poly Tetra fluoro Ethylene (PTFE), Teflon

⑥ $CH_2=\underset{CN}{CH}$ — Acrylonitrile —:,:→ $\{CH_2-\underset{CN}{CH}\}_n$ Poly acrylonitrile / PAN / Orlon

⑦ $CH_2=\underset{O-\overset{O}{\overset{\|}{C}}-CH_3}{CH}$ — Vinyl acetate —:→ $\{CH_2-\underset{O-COMe}{CH}\}_n$ Poly Vinyl acetate / PVA

used in adhesives

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Examples of Polymer:-

Addition

⑧ $n\ CH_2=\underset{\underset{COOEt}{|}}{CH}$ (ethyl acrylate) $\xrightarrow{\text{Polymerisation}}$ $\left[CH_2-\underset{\underset{COOEt}{|}}{CH}-\right]_n$ PEA — Poly ethyl acrylate — **rubber like polymer**

⑨ $CH_2=\underset{\underset{C(=O)-O-CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}$ (Methyl methacrylate) $\xrightarrow{..}$ $\left[CH_2-\underset{\underset{COOMe}{|}}{\overset{\overset{CH_3}{|}}{C}}-\right]_n$ Poly methy meth acrylate (PMMA), Plexiglas, Lucite etc.

transparent like glass, does not break easily; used in lenses, aircraft windows etc.

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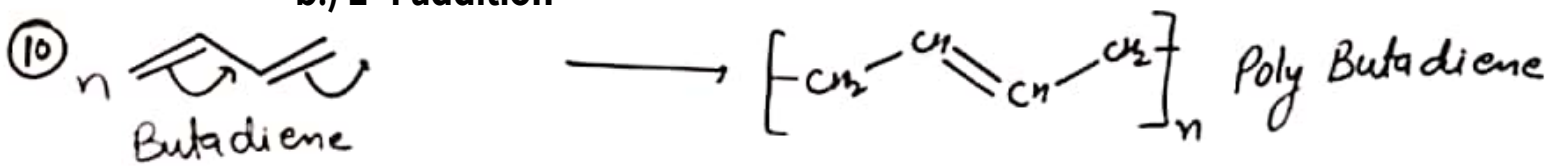
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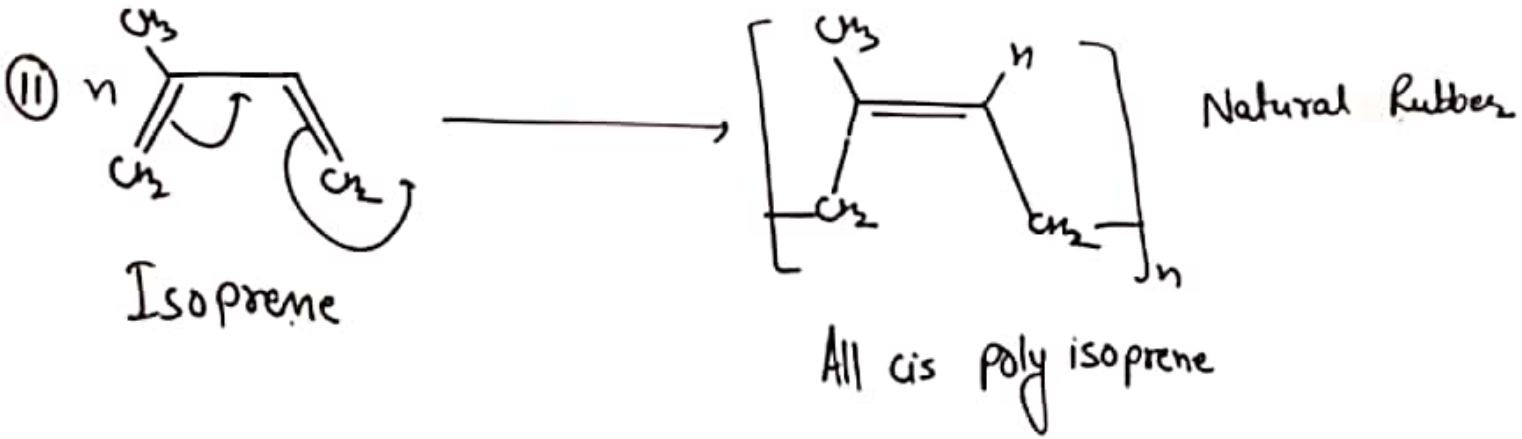
Examples of Polymer:-

Addition

A.] Addition Homopolymers

b.) 1-4 addition

(10) 

(11) 

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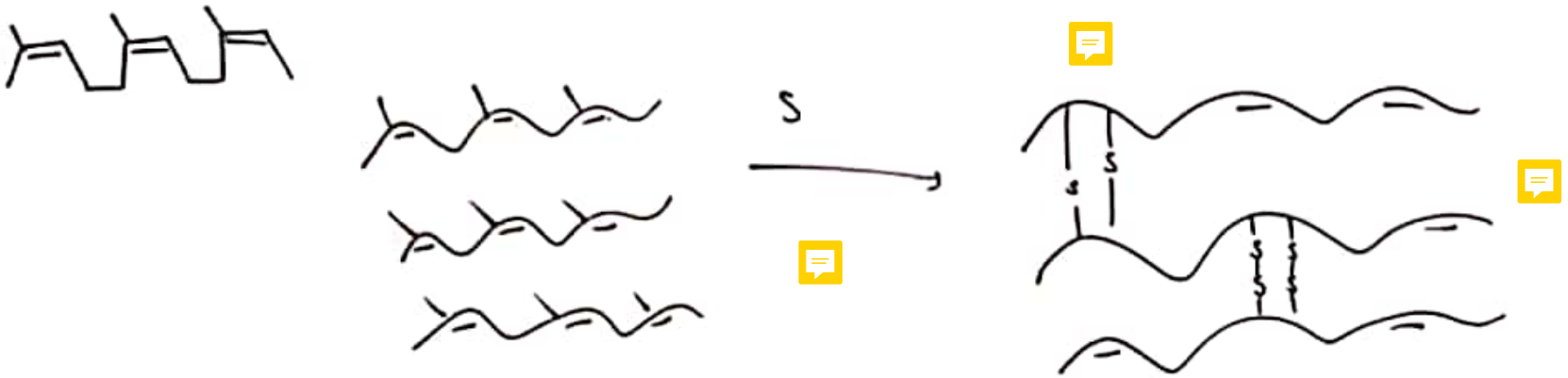
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Examples of Polymer:-

Addition

Vulcanisation :- Natural rubber is not much more elastic and not durable, for improvement of quility of natural rubber is treated with 3-5% Sulpher. This process is called vulcanisation. Vulcanised rubber is relatively more elastic and have high tensile Strength due to formation of cross links of sulpher



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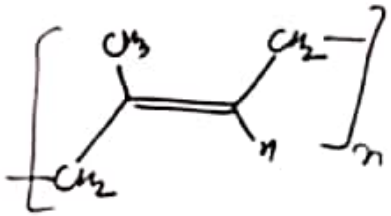
POLYMERS

Examples of Polymer:-

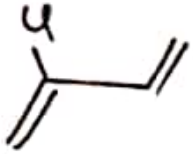
(12) 

Isoprene

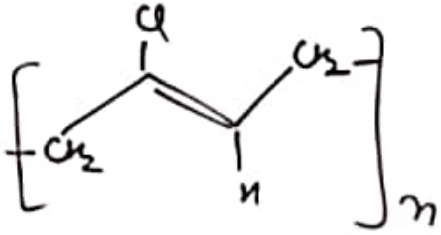
Polymerisation



All Trans poly isoprene
or
Gutta percha rubber

★ (13) 

Chloroprene



all trans poly chloroprene
or
Neoprene
or
Synthetic rubber

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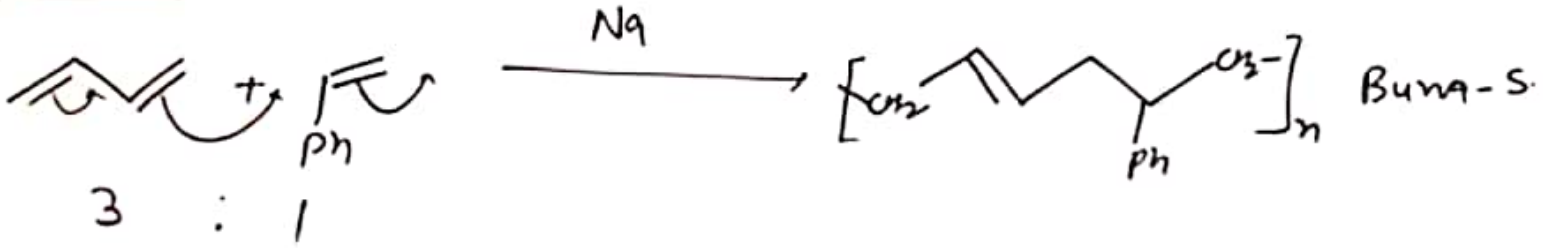
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POLYMERS

Examples of Polymer:-

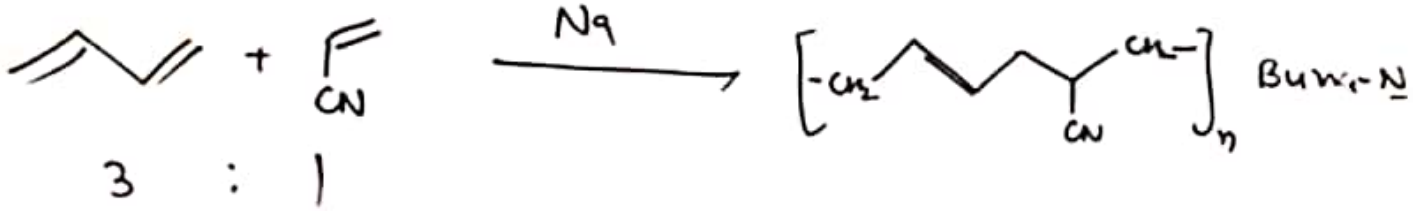
B.] Addition Copolymers

(14) Buna-S

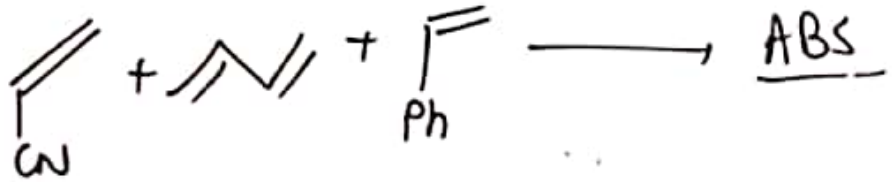


Bu for butadiene
Na for sodium used as catalyst
S for styrene

(15)



(16)



unbreakable plastic

used in car dashboards, bumpers, absorb some amount of shock

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POLYMERS

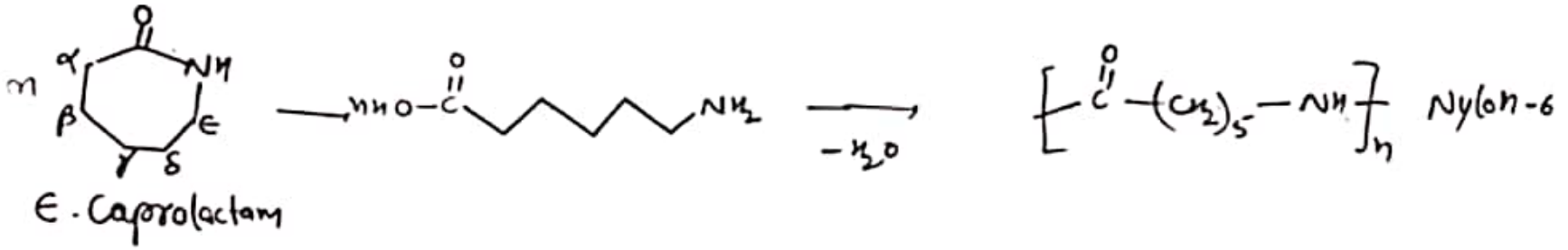
Examples of Polymer:- Condensation polymers.

Nylon chains have intermolecular H-bonding

(A) Polyamides

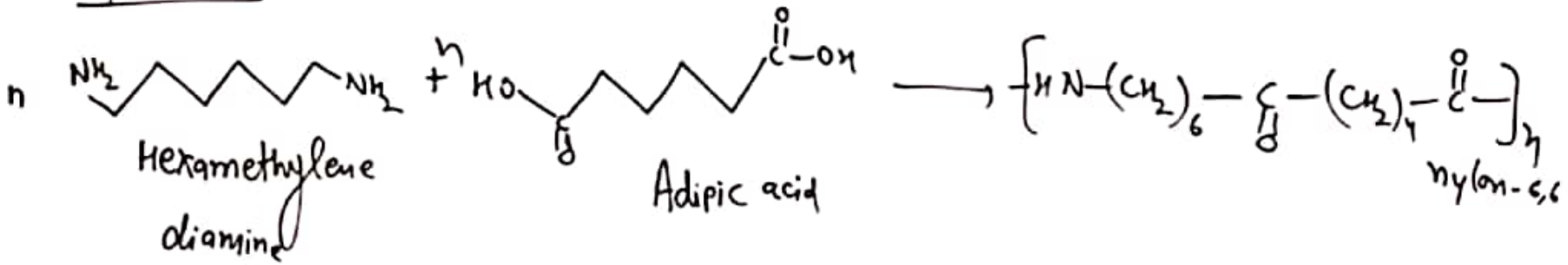
(1) Nylon-6

abbreviated from NewYork London



ε-Caprolactam

(2) Nylon-6,6



3. Nylon 6-10 :
hexamethylene diamine and Sebacic acid

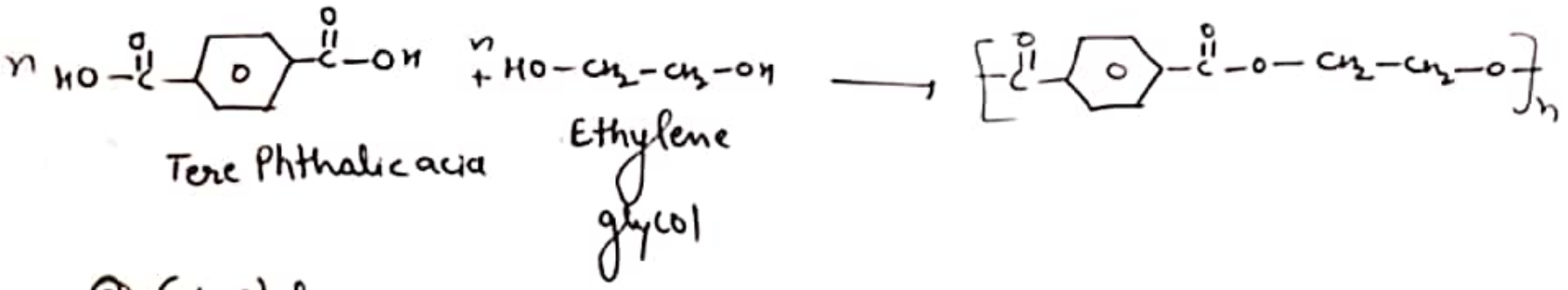
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POLYMERS

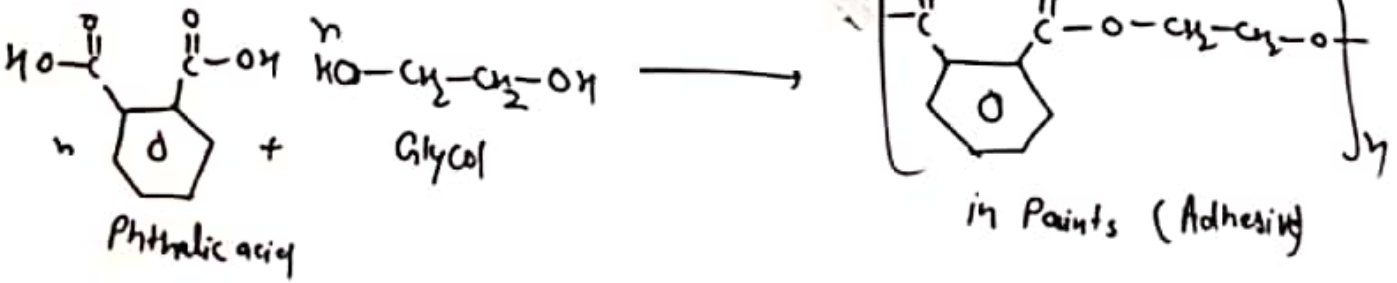
Examples of Polymer:- Condensation polymers.

(B) Poly ester

(1) Terelene = Dacron



(2) Glyptal



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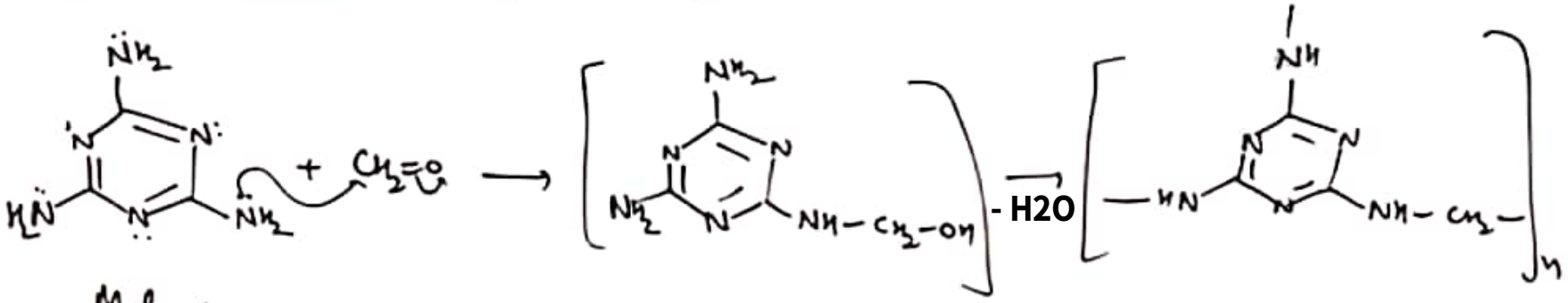
POLYMERS

Examples of Polymer:- Condensation polymers.

© Phenol formaldehyde Polymer
- Linear Polymer (Novalac = Resol)
- 3D Polymer = Bakelite

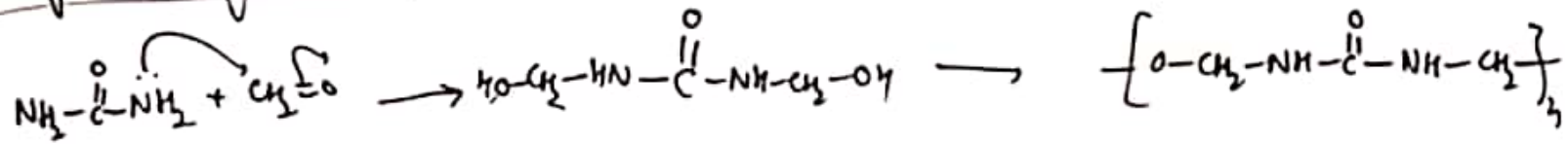
refer phenols

Ⓓ Melmac = Melamine - formaldehyde polymer:-



Melamine

(E) Urea - formaldehyde resin:-



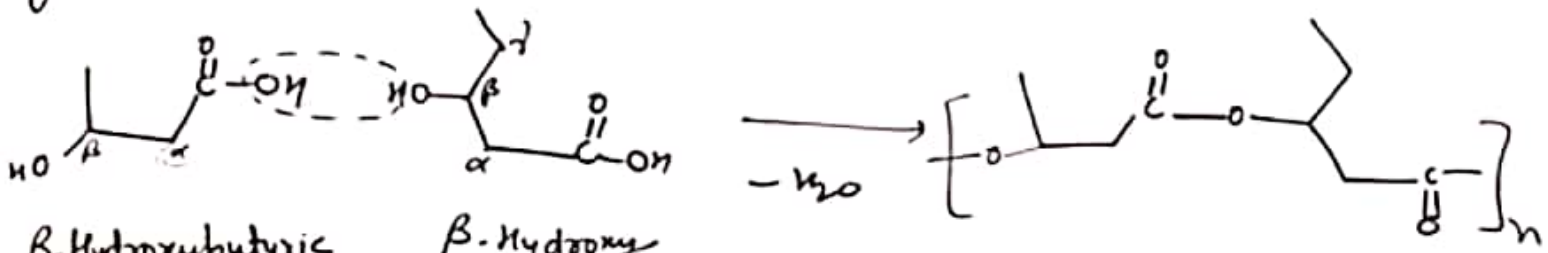
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POLYMERS

Biodegradable Polymers

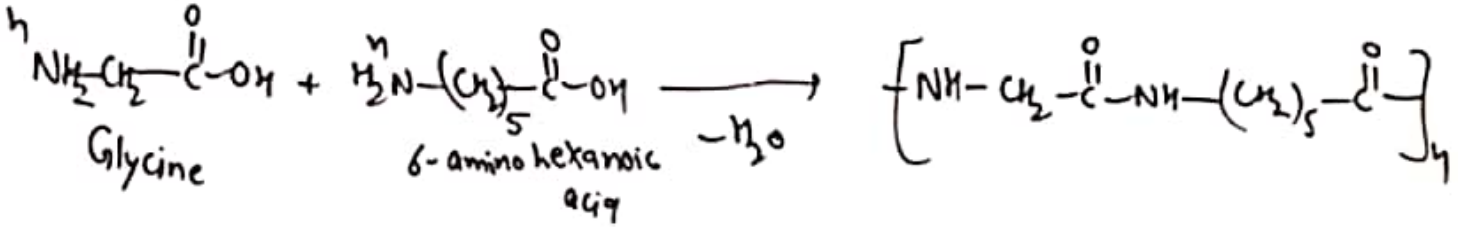
① Poly-β-Hydroxybutyrate-co-β-hydroxy-valarate (PHBV)



β-Hydroxybutyric acid

β-Hydroxy Valaric acid

② Nylon-2-Nylon-6 :-



Glycine

6-amino hexanoic acid

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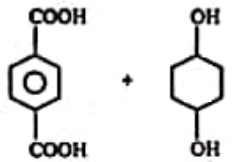
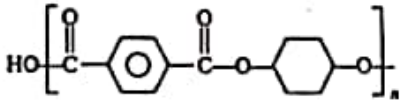
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Some Commercial Polymers

	Monomer	Repeating unit	Polymer name	Uses
(1)	$CH_2=CH_2$ Ethylene	$-CH_2-CH_2-$	Polyethylene	Film, Toys, bottles Plastic bags, sheets, house wares, wire, cable covering, and shipping containers.
(2)	$CH_2=CH-Cl$ Vinyl chloride	$-CH_2-\underset{Cl}{CH}-$	Polyvinyl chloride (PVC)	Squeez bottles, pipe, siding, flooring, films, sheets, records, hand bags.
(3)	$CH_3-CH=CH_2$ Propene	$-CH_2-\underset{CH_3}{CH}-$	Polypropene	Moulded caps, margarine tubes, indoor/outdoor carpeting
(4)	$C_6H_5-CH=CH_2$ Styrene	$-CH_2-\underset{C_6H_5}{CH}-$	Polystyrene	Packing, toys, clear cups, egg cartoons, hot drink cups.
(5)	$CF_2=CF_2$ (Tetrafluoroethylene)	$-CF_2-CF_2-$	Polytetrafluoro ethylene (PTFE) Teflon	Non-sticking surfaces, liners, cable insulation.
(6)	$CH_2=CH-CN$ Vinyl cyanide	$-CH_2-\underset{CN}{CH}-$	Polyvinylcyanide, polyacronitrite, orlon or acrilan	Rugs, blanket, yarn simulated fur.
(7)	$CH_2=\underset{CH_3}{C}-\overset{O}{\overset{\|\|}{C}}-O-CH_3$ Methyl methacrylate	$-CH_2-\underset{CH_3}{\overset{COOCH_3}{C}}-$	Polymethyl methacrylate] (PMMA) plexiglas, lucite	Lighting fixtures, solar panels, skylights.
(8)	$CH_2=CH-O-\overset{O}{\overset{\|\|}{C}}-CH_3$ Vinyl acetate	$-CH_2-\underset{OCOCH_3}{CH}-$	Polyvinylacetate	Latex paints, adhesives
(9)	$CH_2=CH-CH=CH_2$ 1,3-Butadiene	$-CH_2-CH=CH-CH_2-$	Polybutadiene (Buna rubber)	In the manufacture of tyres, hoses etc.

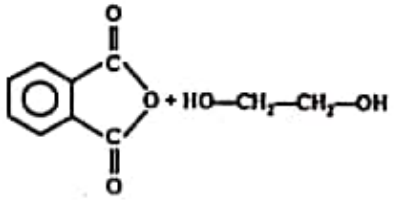
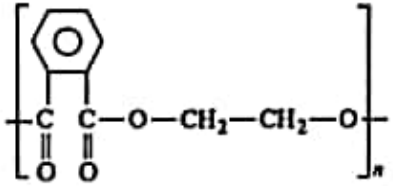
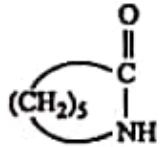
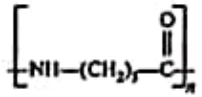
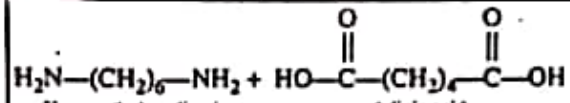
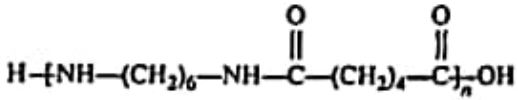
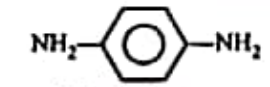
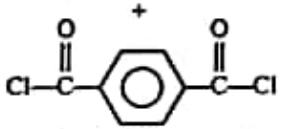
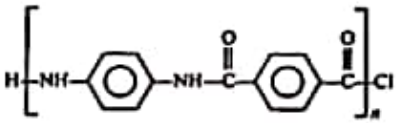
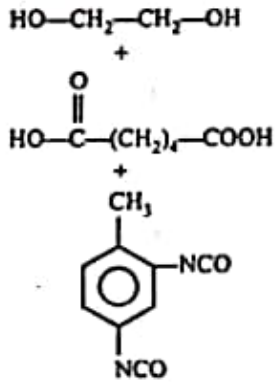
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	Monomer	Repeating unit	Polymer name	Uses
(10)	$CH_2=CH-Cl$ (vinyl chloride) + $CH_2=CCl_2$ (vinylidene chloride)	$-CH_2-\underset{\displaystyle Cl}{\underset{\vert}{CH}}-CCl_2-CH_2-$	Saran	Film for wrapping food
(11)	$C_6H_5-CH=CH_2$ (styrene) + $CH_2=CH-CN$ (acrylonitrile)	$-CH_2-\underset{\displaystyle C_6H_5}{\underset{\vert}{CH}}-\underset{\displaystyle CN}{\underset{\vert}{CH}}-CH_2-$	SAN	Dish washer, vaccum cleaner parts
(12)	$CH_2=CH-CN + CH_2=CH-CH=CH_2$ + $C_6H_5-CH=CH_2$	—	ABS	Bumpers, helmets, telephones, luggage
(13)	$CH_2=\underset{\displaystyle CH_3}{\underset{\vert}{C}}-CH_3 + CH_2=\underset{\displaystyle CH_3}{\underset{\vert}{C}}-CH=CH_2$	—	Butyl rubber	Innertubes, balls, inflatable sporting goods.
(14)	$C_6H_5-CH=CH_2 + CH_2=CH-CH=CH_2$	—	Buna-S, SBR	Tyres and mechanical rubber goods
(15)	$CH_2=CH-CN + C_6H_5-CH=CH_2$	$-CH_2-\underset{\displaystyle CN}{\underset{\vert}{CH}}-CH_2-\underset{\displaystyle C_6H_5}{\underset{\vert}{CH}}-$	Buna-N, NBR	Rubber resistant to oils and chemicals
(16)	$CH_2=\underset{\displaystyle Cl}{\underset{\vert}{C}}-CH=CH_2$	$-CH_2-\underset{\displaystyle Cl}{\underset{\vert}{C}}=CH-CH_2-$	Neoprene	Rubber in gasoline pump hoses and rubber tubing in automobile engines
(17)	HOOC–C$_6$H$_4$–COOH + $HO-CH_2-CH_2-OH$	$HO-\left[C(=O)-C_6H_4-C(=O)-O-CH_2-CH_2-O\right]_n$	Poly (ethylene terephthalate), Terylene, Dacron or Mylar	Chothing, Sails
(18)			Kodel polyester	Fibres and films

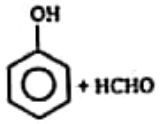
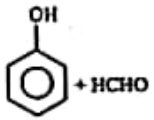
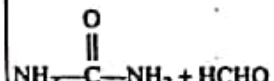
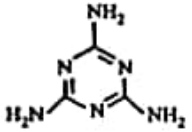
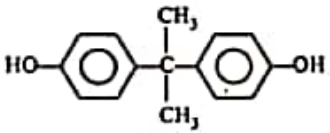
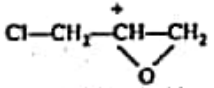
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	Monomer	Repeating unit	Polymer name	Uses
(19)			Polyethylene phthalate, Alkyd resin (Glyptal)	Used in paints and lacquers
(20)	 Caprolactum		Nylon-6	Fibre
(21)	 Hexamethylenediamine Adipic acid		Nylon-66	As fibres
(22)	 1, 4-Diaminobenzene +  Terephthaloyl chloride		Kevlar	Light weight personal body armour (bullet-resistant vests)
(23)		—	Polyurethane foam	Flexible foams

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	Monomer	Repeating unit	Polymer name	Uses
(24)	 + HCHO Ratio P/F = 0.8	—	Bakelite or resol	Adhesive binders and laminates
(25)	 + HCHO Ratio P/F = 0.25	—	Novolak resin	Radio cabinet, camera cases, telephone parts, phonograph records
(26)		—	Urea-formaldehyde resin	Buttons, bottle caps, surgical items
(27)	Melamine + HCHO  + HCHO	—	Melamine-formaldehyde resin	Manufacture of kichenware, used as adhesive
(28)	 *Bis* phenol +  Epichlorohydrin	—	Epoxy resin	Strong adhesive properties to bind glass, porcelain, metal and wood

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